

SAVE-MAP

December 3, 2025

```
[1]: %reload_ext autoreload
    %autoreload 2

[2]: from waymo_agent import *

[3]: from waymo_agent.osmnx.inspect import *
    from waymo_agent.osmnx.osmnx_constants import *
    from waymo_agent.osmnx.sparsify import *
    from waymo_agent.osmnx.utils import _max_speed_int

[4]: G_raw = ox.graph_from_place(
    "Manhattan, New York, USA",
    network_type="drive",
    simplify=True,
)

print(f"Nodes: {G_raw.number_of_nodes()}")
print(f"Edges: {G_raw.number_of_edges()}")
```

Nodes: 4619

Edges: 9901

0.1 Basic Save / Load

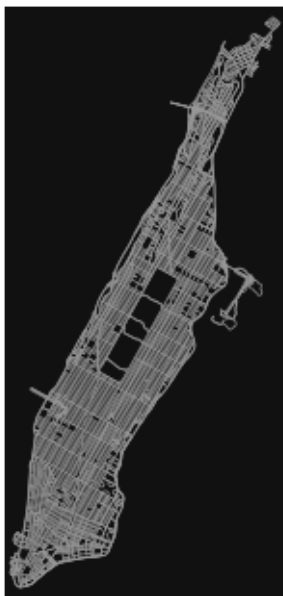
```
[5]: MAP_DIR

[5]: PosixPath('/Users/Akseldkw/coding/Columbia/RL-Project/maps')

[6]: ox.save_graphml(G_raw, filepath=MAP_DIR / MANHATTAN_RAW_GRAPH)

[7]: G_out = ox.load_graphml(filepath=MAP_DIR / MANHATTAN_RAW_GRAPH)

[8]: fig, ax = ox.plot_graph(G_out, node_size=0, edge_linewidth=0.5, figsize=(4, 4))
```



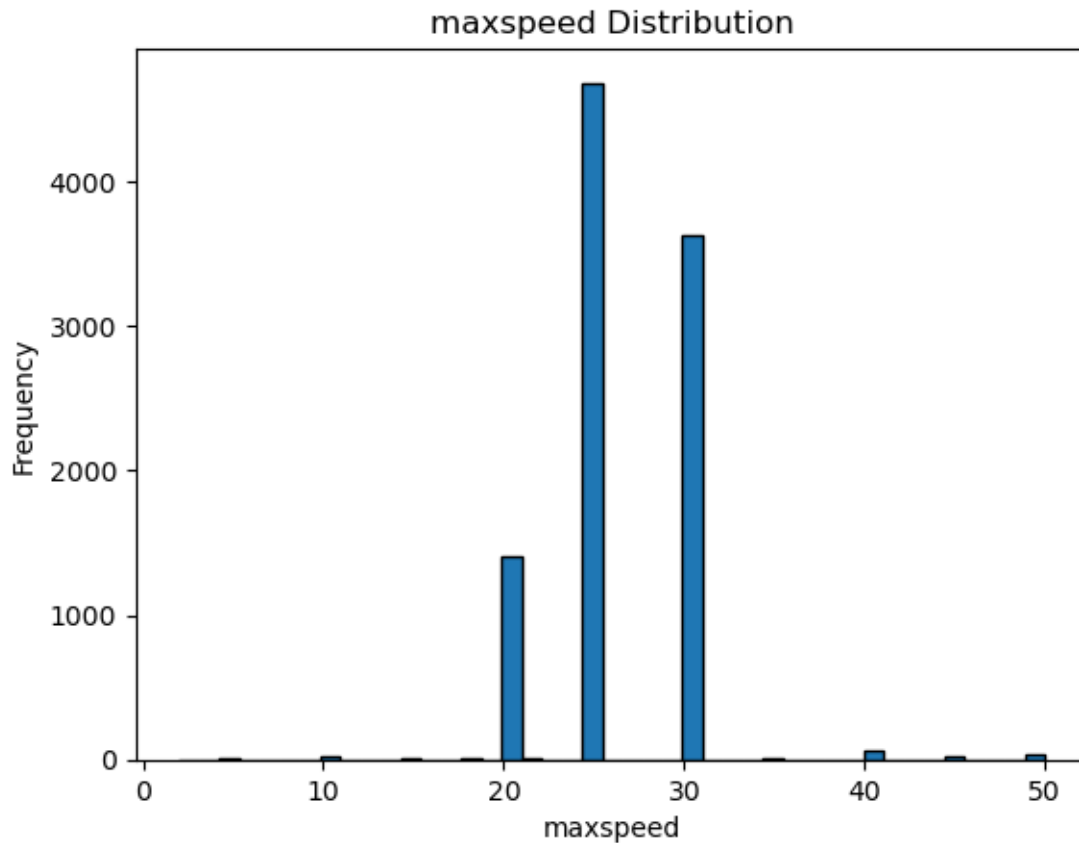
0.2 Preprocess Graph

```
[9]: G = preprocess_graph(G_raw)
```

```
[10]: collect_unique_graph_attr(G, "maxspeed")
```

```
[10]: {2, 5, 10, 14, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40, 44, 45, 48, 50}
```

```
[11]: plot_maxspeed(G)
```



```
[12]: collect_unique_graph_attr(G, "maxspeed")
```

```
[12]: {2, 5, 10, 14, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40, 44, 45, 48, 50}
```

```
[13]: ox.save_graphml(G, filepath=MAP_DIR / MANHATTAN_PROCESSED_GRAPH)
```

0.3 Sparsify Graph

```
[14]: G = sparsify_graph(G)
```

Removing 4727 non-major edges...

Graph after edge removal: G_sparsified.number_of_nodes()=3517,

G_sparsified.number_of_edges()=5173

Cleaning all edge attributes before simplification...

Cleaning all edge attributes before simplification...

Removing 110 isolated or pass-through nodes...

Cleaning all edge attributes before simplification...

Final simplified graph: G_ret.number_of_nodes()=774,

G_ret.number_of_edges()=1613

```
[15]: sparse_graph_name = MANHATTAN_SPARSE_GRAPH.format(G.number_of_nodes())
sparse_graph_name
```

```
[15]: 'manhattan-sparse-774-nodes.graphml'
```

```
[16]: ox.save_graphml(G, filepath=MAP_DIR / sparse_graph_name)
```

Load Sparse Graph

```
[17]: G = ox.load_graphml(filepath=MAP_DIR / sparse_graph_name)
```

```
[18]: edge_colors, edge_widths = OSMNXConstants.get_edge_colors(G)
fig, ax = ox.plot_graph(
    G,
    edge_color=edge_colors,
    edge_linewidth=edge_widths,
    **{**DEFAULT_OX_PLOT_NOTEBOOK | {"figsize": (4, 4)}},
)
```



0.4 Enrich Graph

Add lambda (demand) and charger info

```
[19]: G = ox.load_graphml(filepath=MAP_DIR / sparse_graph_name)
post_load(G)
```

```
[20]: enrich_graph(G)
```

Assigned lambda values to nodes. Total lambda: 8.8753 (target: 7.7400)

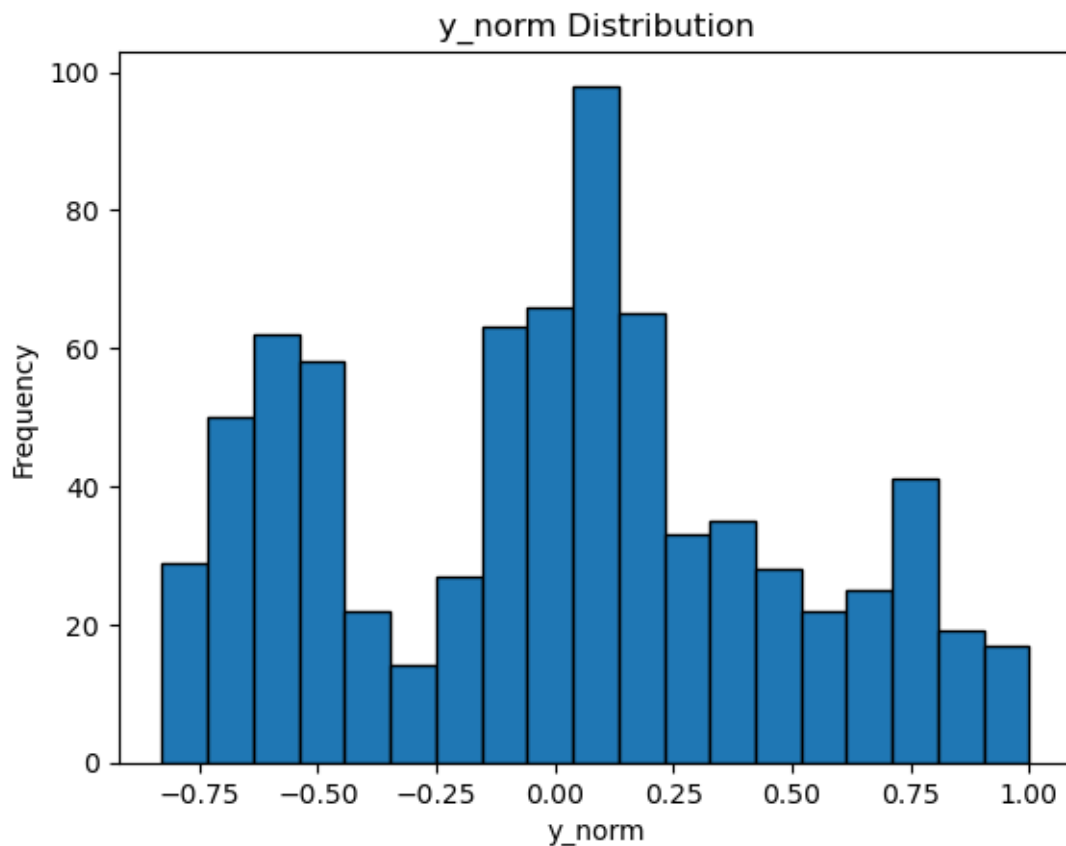
```
[21]: inspect_graph(G)
```

Graph has 774 nodes and 1613 edges.

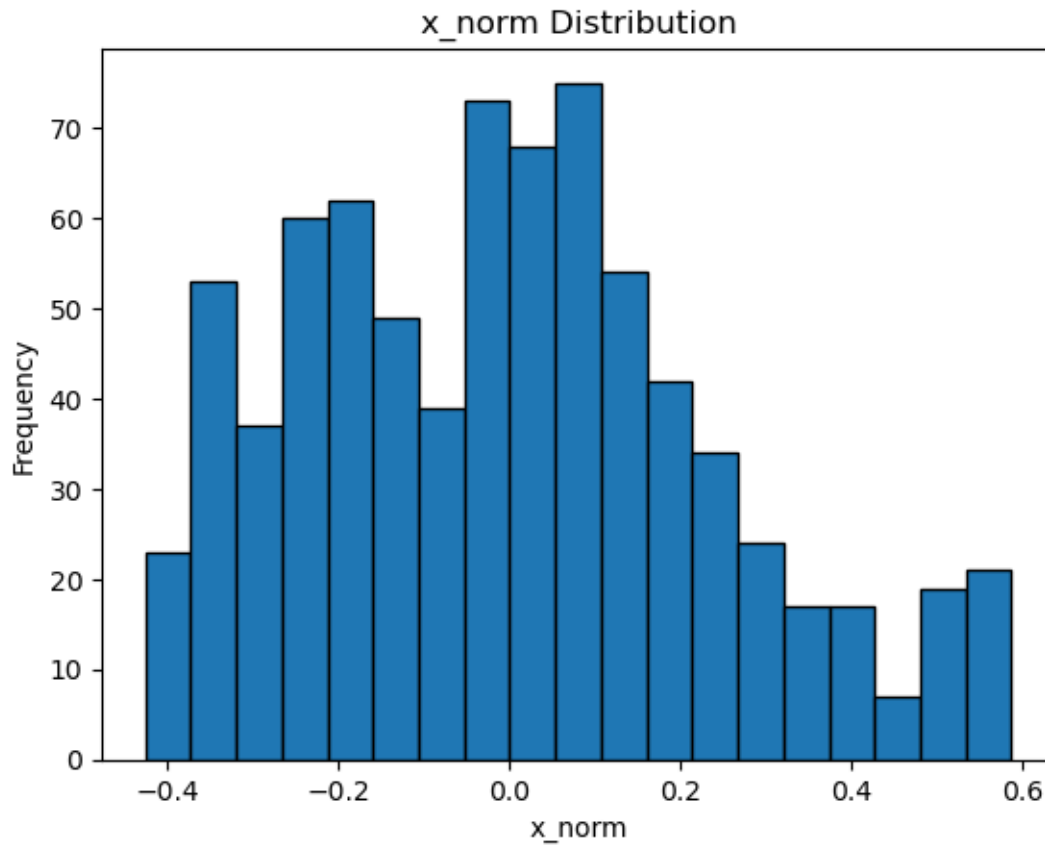
Sample node -sample_node_id=42431004- attributes: {'y': 40.7798127, 'x': -73.9844806, 'highway': 'traffic_signals', 'street_count': 4, 'lambda': 0.025363779161605114, 'x_centered': -444.83847409591544, 'y_centered': 2774.091377252713, 'x_norm': -0.06211665299714188, 'y_norm': 0.38737043106126506}

Sample edge attributes: {'osmid': 675967321, 'highway': 'primary', 'lanes': "('6', '5')", 'maxspeed': 25.0, 'name': 'Bowery', 'oneway': False, 'reversed': False, 'length': 225.74104147101343, 'geometry': <LINESTRING (-73.995 40.718, -73.995 40.717, -73.996 40.717)>, 'travel_time_min': 0.5417784995304322}

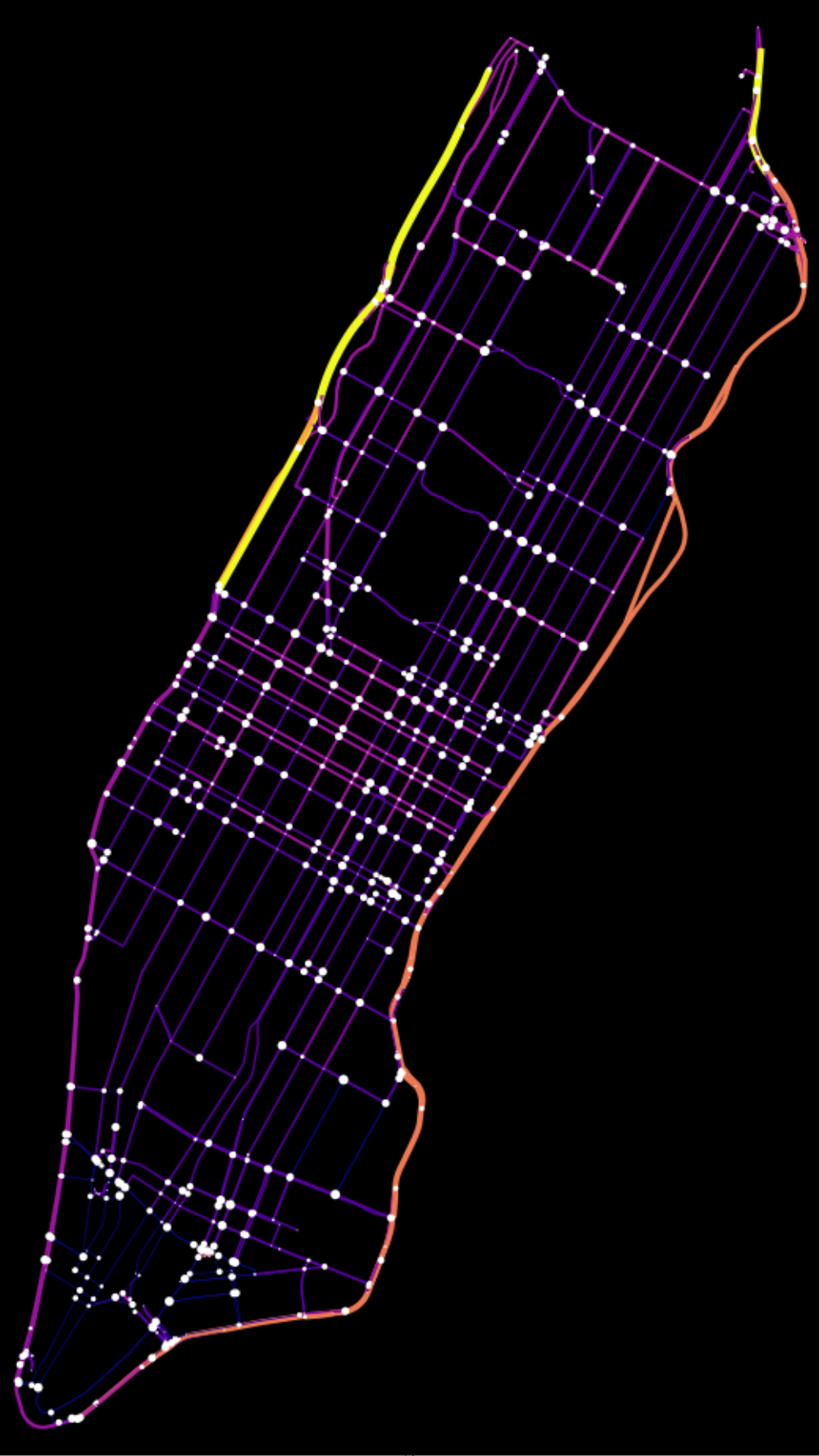
```
[25]: hist_edge_or_node_attr(G, "y_norm", "node")
```



```
[26]: hist_edge_or_node_attr(G, "x_norm", "node")
```

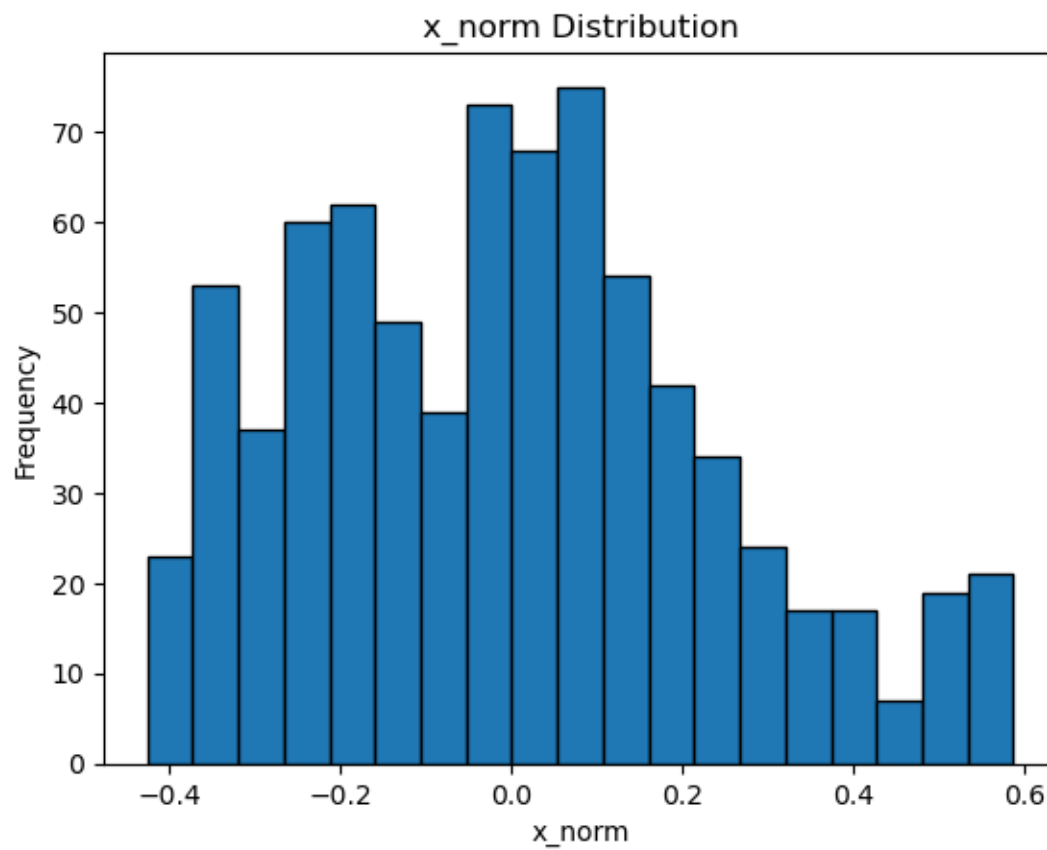


```
[27]: edge_colors, edge_widths = get_edge_color_by_speed(G)
node_colors, node_sizes = get_node_colors_and_sizes(G)
fig, ax = ox.plot_graph(
    G,
    edge_color=edge_colors,
    edge_linewidth=edge_widths,
    node_size=node_sizes,
    node_color=node_colors,
    figsize=(12, 12),
    show=False,
    bgcolor="black",
)
```



```
[28]: name = MANHATTAN_ENRICHED_GRAPH.format(G.number_of_nodes())  
      ox.save_graphml(G, filepath=MAP_DIR / name)
```

```
[29]: hist_edge_or_node_attr(G, "x_norm", "node")
```



```
[ ]:
```